Student Performance Analytics Dashboard: A Machine Learning-Based Web System for Score Prediction

M. Akhila Devi, Sk. Harshitha, I. Keerthana, and B. Anupama
Department of Computer Science and Engineering, KLH CSE Bowrampet Campus, India
Guide: Ms. P. Prathusha, Assistant Professor

Abstract— Student performance is associated with academic and learning-related factors such as study hours, attendance, previous scores, assignment performance and classroom participation. This paper presents a Student Performance Analytics Dashboard that combines exploratory data analysis, Random Forest Regression and a web interface for score prediction. The supplied dataset contains 800 synthetically generated student records and 13 columns, including an identifier, 11 predictor attributes and the final-score target. Categorical attributes are one-hot encoded and a Random Forest Regressor with 100 trees is trained using an 80:20 train-test split. In a reproducible run using a fixed random seed of 42, the model achieved MAE of 4.30, RMSE of 5.53 and R^2 of 0.594 on the held-out test set. A Flask-based interface accepts student attributes and displays a predicted score and performance category, while an analytics view presents score patterns and feature importance. The system is intended as an academic decision-support prototype; its synthetic dataset limits direct generalization to real student populations.

Index Terms— educational data mining, student performance, Random Forest Regression, feature importance, Flask, academic analytics.

I. INTRODUCTION

Educational institutions maintain records containing academic results, attendance and learning behaviours. Reviewing many records manually can make it difficult to identify patterns and provide timely academic support. Data science methods can summarize these records, while supervised machine learning can estimate a numerical outcome from multiple input variables. The proposed Student Performance Analytics Dashboard combines these capabilities in a lightweight web application. Its workflow covers dataset preparation, exploratory analysis, regression-based prediction and visualization.

The system is designed to support interpretation rather than replace instructor judgement. A predicted score is an estimate based on patterns in the supplied dataset and should not be treated as a definitive assessment of an individual student.

II. RELATED WORK

Educational data mining has been used to study student achievement and to compare predictive algorithms. Ya■c■ examined machine-learning approaches for predicting university students' academic performance [1]. Tomasevic *et al.* compared supervised data-mining techniques for student exam performance [2]. Nachouki *et al.* used Random Forest and predictor-importance analysis to study course-grade prediction [3]. These works motivate the use of supervised learning and feature analysis for educational data.

Random Forest, introduced by Breiman, combines multiple decision trees to improve predictive robustness [4]. Romero and Ventura reviewed educational data mining techniques and their applications [5]. The project gap identified from the review is the need for a single, accessible workflow that combines data exploration, student-level score prediction and understandable visual summaries.

III. DATASET AND PREPROCESSING

The supplied CSV contains 800 rows and 13 columns. Student_ID is used only as an identifier and is excluded from model training. The 11 input features are Study_Hours_per_Week, Attendance_Percentage, Previous_Score, Assignment_Score, Class_Participation, Sleep_Hours_per_Night, Tutoring_Sessions, Extracurricular_Activities, Internet_Access, Motivation_Level and Parental_Involvement. Final_Score is the numerical target.

A direct inspection of the supplied file found no missing values and no duplicate rows. Numeric columns are retained as numeric predictors. The four categorical inputs are converted using one-hot encoding, with unseen categories handled by the preprocessing pipeline. The dataset is synthetically generated for project experimentation; it is not a verified sample of real institutional records.

IV. PROPOSED METHODOLOGY

The dataset is divided into training and test subsets using an 80:20 split and random_state=42. A preprocessing stage applies one-hot encoding to categorical variables and passes numeric variables through unchanged. A RandomForestRegressor with n_estimators=100 and random_state=42 is then fitted on the training data. The fitted pipeline generates predictions for the held-out test records.

The evaluation uses Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and coefficient of determination (R^2). MAE measures average absolute prediction error, RMSE penalizes larger errors more strongly, and R^2 summarizes the proportion of target variance explained by the model relative to a mean-baseline predictor.

V. SYSTEM IMPLEMENTATION

The web application is implemented with Flask. The prediction page collects the 11 student attributes through numeric fields and categorical selectors. On submission, the application passes the values to the trained model and presents a predicted final score together with a performance label. The supplied interface screenshot shows a prediction of 42.73 and the label "Low Performance"; this is an illustrative application output and is not the test-set metric reported in this paper.

The analytics interface presents a study-hours-versus-final-score scatter plot, a score-distribution chart and a feature-importance chart. Chart.js is identified in the project presentation as the visualization library. Joblib is listed for model storage. The project presentation describes local execution through the Flask development server.

VI. EXPERIMENTAL RESULTS

On the supplied 800-record dataset, the fixed-seed evaluation produced MAE = 4.296, RMSE = 5.533 and R^2 = 0.594. The mean final score in the complete dataset is 51.32, with a standard deviation of 8.93 and observed range 22.5–80.9.

Aggregated feature importance from the fitted Random Forest ranks Previous Score first (0.348), followed by Study Hours per Week (0.241), Class Participation (0.136), Attendance Percentage (0.066) and Assignment Score (0.060). Feature importance indicates how the fitted model uses predictors; it does not establish causal influence.

VII. OUTPUTS AND DISCUSSION

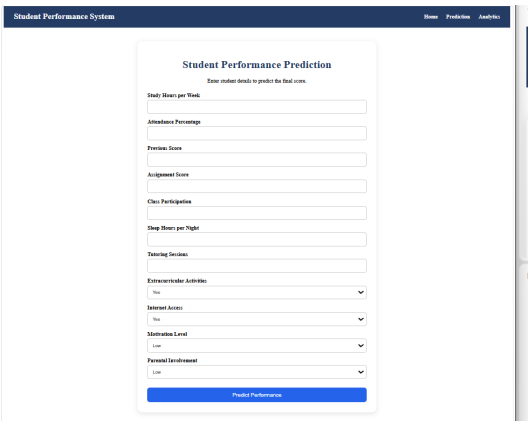


Fig. 1. Student input form for score prediction.

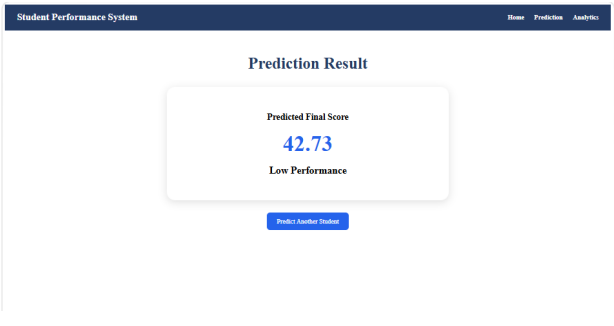


Fig. 2. Example prediction result displayed by the application.

The form supports entry of study hours, attendance, previous score, assignment score, participation, sleep, tutoring and categorical learning-context attributes. The result page converts the model output into a simple score and performance label. This design makes the prediction workflow accessible to users without requiring them to run model code directly.

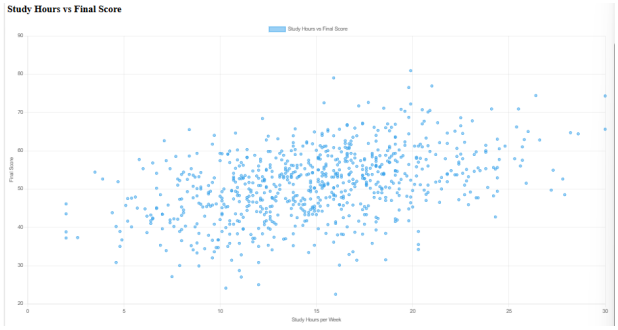


Fig. 3. Study hours per week versus final score.

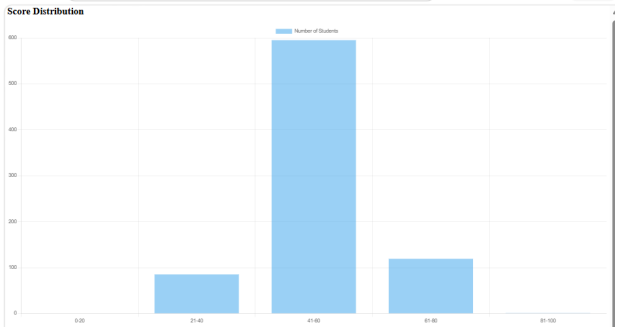


Fig. 4. Distribution of final scores in the dashboard.

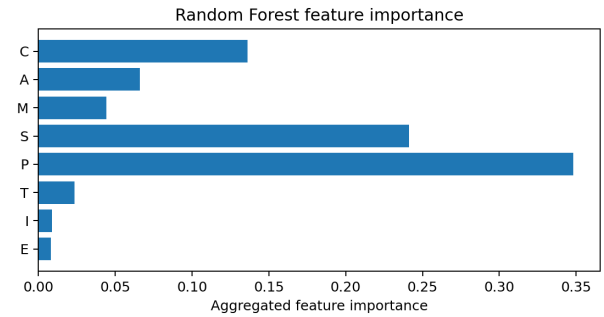


Fig. 5. Feature importance aggregated to the original input attributes for the reproduced model.

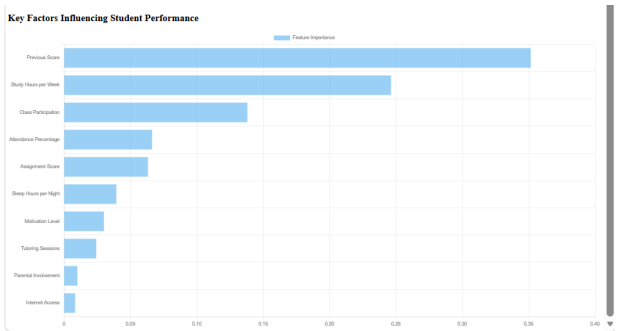


Fig. 6. Feature-importance visualization captured from the project dashboard.

VIII. LIMITATIONS AND FUTURE WORK

The dataset is synthetic, so model performance on real student populations has not been established. Only one regression algorithm is included in the current implementation, and the reported evaluation uses one fixed train-test split. The web output shown in the screenshots is not sufficient by itself to establish model accuracy. Future work should evaluate real, ethically sourced institutional data; use cross-validation; compare Random Forest with linear regression, decision-tree and gradient-boosting baselines; tune hyperparameters; and report uncertainty and subgroup performance. The interface can also be extended with separate student and faculty views and secure deployment.

IX. CONCLUSION

This paper presented a student performance analytics prototype that integrates data preprocessing, Random Forest score prediction and a Flask dashboard. On the supplied synthetic dataset, a reproducible 80:20 evaluation achieved MAE of 4.30, RMSE of 5.53 and R^2 of 0.594. Previous Score and Study Hours per Week were the two most important predictors in the fitted model. The system demonstrates an end-to-end analytics workflow, while real-world validation remains necessary before operational use.

REFERENCES

[1] M. Ya■c■, "Educational data mining: Prediction of students' academic performance using machine learning algorithms," *Smart Learning Environments*, vol. 9, art. no. 11, 2022, doi: 10.1186/s40561-022-00192-z.

[2] N. Tomasevic, N. Gvozdenovic, and S. Vranes, "An overview and comparison of supervised data mining techniques for student exam performance prediction," *Computers & Education*, vol. 143, art. no. 103676, 2020, doi: 10.1016/j.compedu.2019.103676.

[3] M. Nachouki, E. A. Mohamed, R. Mehdi, and M. Abou Naaj, "Student course grade prediction using the random forest algorithm: Analysis of predictors' importance," *Trends in Neuroscience and Education*, vol. 33, art. no. 100214, 2023, doi: 10.1016/j.tine.2023.100214.

[4] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[5] C. Romero and S. Ventura, "Educational data mining: A review of the state of the art," *IEEE Transactions on Systems, Man, and Cybernetics, Part C: Applications and Reviews*, vol. 40, no. 6, pp. 601–618, 2010, doi: 10.1109/TSMCC.2010.2053532.

APPENDIX A — IMPLEMENTATION OUTPUTS

The following figures are the application and analytics outputs supplied with the project materials. They document the implemented user workflow and the dashboard visualizations.

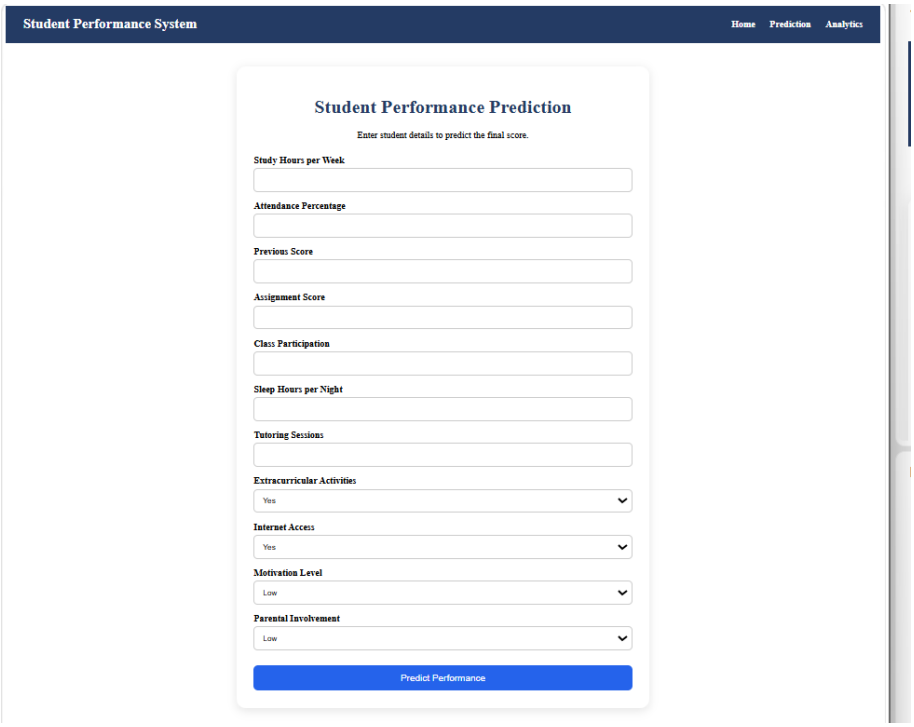


Fig. A1. Student performance prediction form with academic and behavioural inputs.

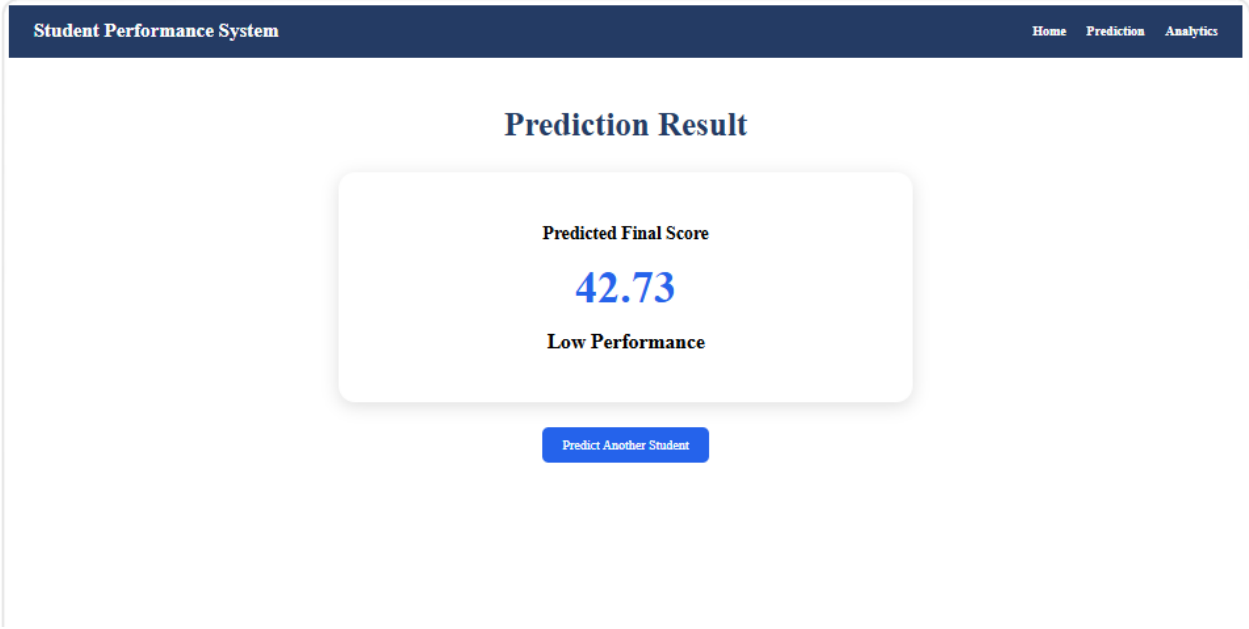


Fig. A2. Example prediction result: 42.73, classified as Low Performance.

APPENDIX B — ANALYTICS AND EXPERIMENTAL DETAILS

The dashboard plots the relationship between weekly study hours and final score, the distribution of final scores, and the relative importance of input features. The first two charts below are reproduced from the supplied application screenshots. The final chart is generated from the fixed-seed model evaluation described in the paper.

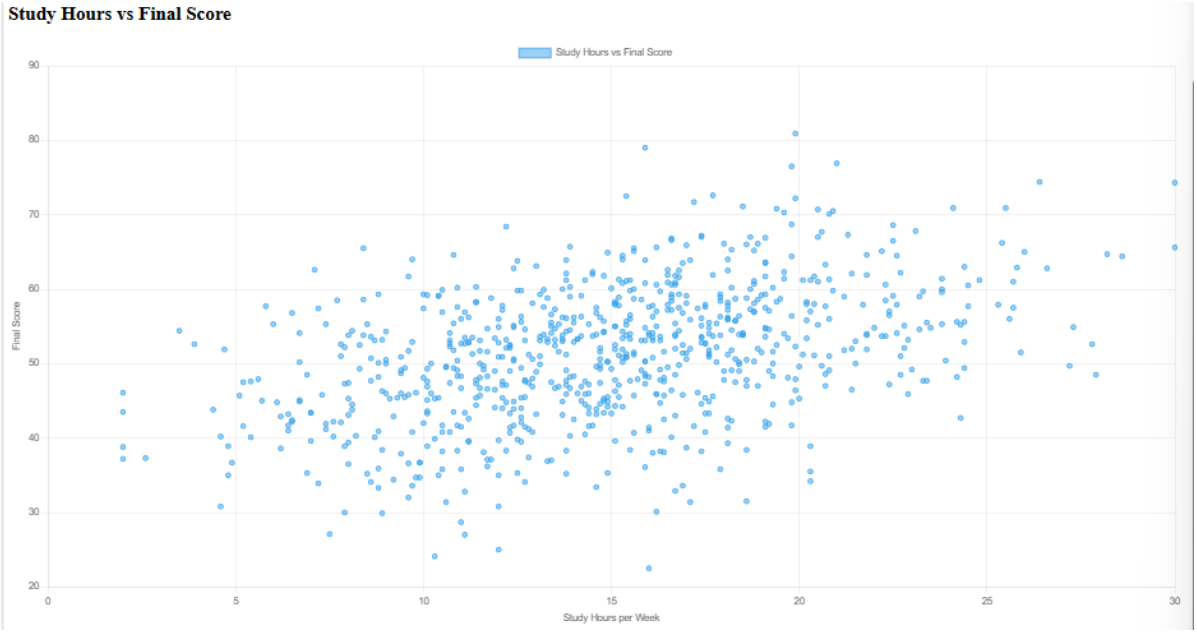


Fig. B1. Study hours per week versus final score.

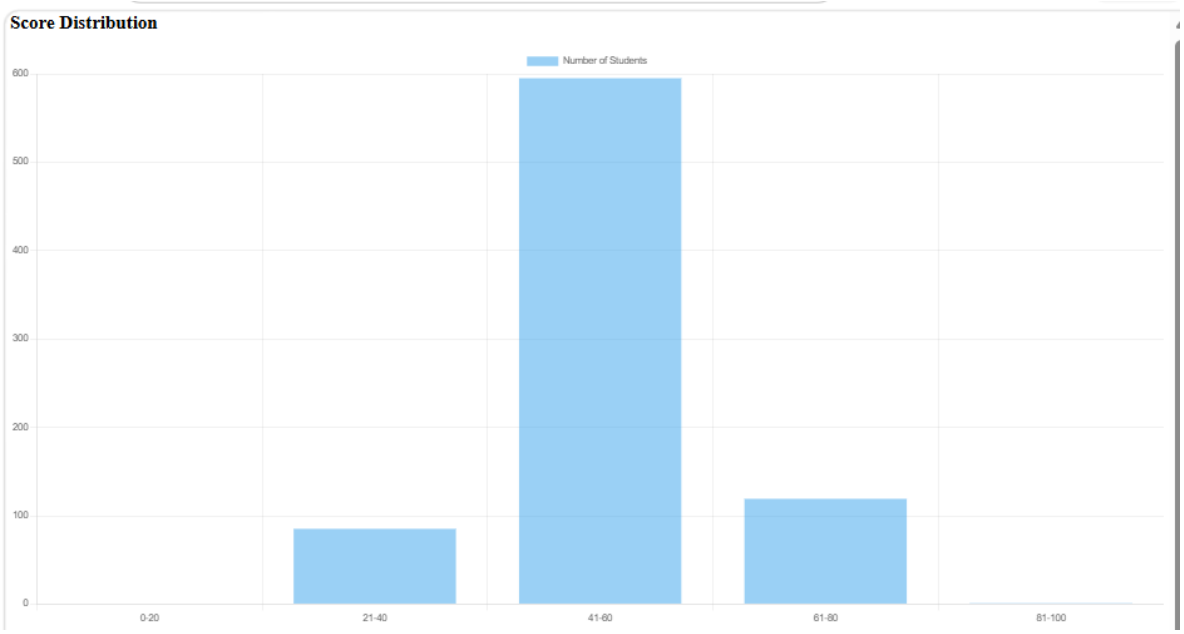


Fig. B2. Distribution of student final scores.

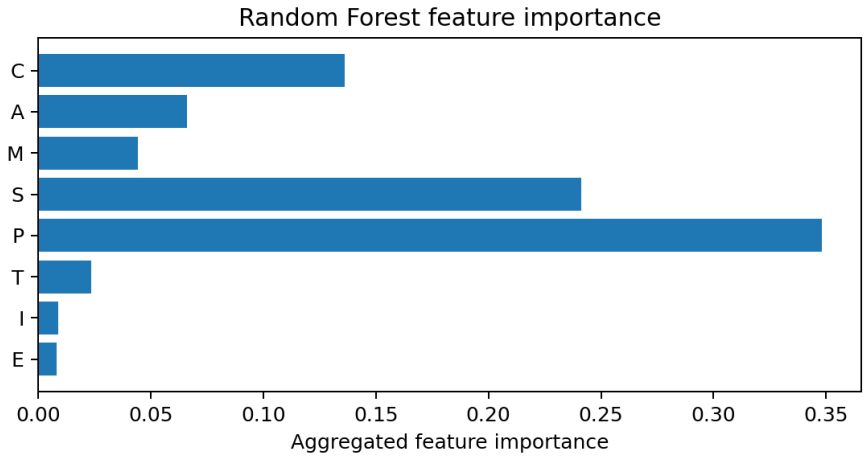


Fig. B3. Feature importance aggregated to the original predictor attributes.

Reproduced evaluation setup: 800 records; 11 predictors; target = Final_Score; 80:20 train-test split; random_state = 42; RandomForestRegressor(n_estimators=100, random_state=42); MAE = 4.296; RMSE = 5.533; R^2 = 0.594.